

MA6.301: Probability and Statistics

Quiz 1 - Solutions and Grading Rubric

Solutions

Problem 1 [2 Points]

Statement: Let $X_1, \dots, X_n$ be independent random variables having a common PMF with mean μ and variance σ^2 . Let $Y = \frac{1}{n}(X_1 + X_2 + \dots + X_n)$. Show that $\mathbb{E} \left[\sum_{i=1}^n (X_i - Y)^2 \right] = (n-1)\sigma^2$.

Solution: First, expand the algebraic expression inside the expectation:

$$\sum_{i=1}^n (X_i - Y)^2 = \sum_{i=1}^n (X_i^2 - 2X_iY + Y^2)$$

Distribute the summation:

$$\sum_{i=1}^n (X_i - Y)^2 = \sum_{i=1}^n X_i^2 - 2Y \sum_{i=1}^n X_i + \sum_{i=1}^n Y^2$$

Since $Y = \frac{1}{n} \sum_{i=1}^n X_i$, we know that $\sum_{i=1}^n X_i = nY$. Also, summing the constant Y^2 exactly n times yields nY^2 :

$$\begin{aligned} \sum_{i=1}^n (X_i - Y)^2 &= \sum_{i=1}^n X_i^2 - 2Y(nY) + nY^2 \\ \sum_{i=1}^n (X_i - Y)^2 &= \sum_{i=1}^n X_i^2 - 2nY^2 + nY^2 = \sum_{i=1}^n X_i^2 - nY^2 \end{aligned}$$

Next, apply the expectation operator to both sides:

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - Y)^2 \right] = \sum_{i=1}^n \mathbb{E}[X_i^2] - n\mathbb{E}[Y^2]$$

Using the variance formula $\mathbb{E}[Z^2] = \text{Var}(Z) + (\mathbb{E}[Z])^2$, we find the second moments for X_i and Y : For X_i : $\mathbb{E}[X_i^2] = \sigma^2 + \mu^2$

For Y : The mean is $\mathbb{E}[Y] = \mu$. Because the variables are independent, the variance is $\text{Var}(Y) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$. Thus, $\mathbb{E}[Y^2] = \frac{\sigma^2}{n} + \mu^2$.

Substitute these back into the expectation equation:

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - Y)^2 \right] = \sum_{i=1}^n (\sigma^2 + \mu^2) - n \left(\frac{\sigma^2}{n} + \mu^2 \right)$$

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - Y)^2 \right] = n(\sigma^2 + \mu^2) - \sigma^2 - n\mu^2$$

$$\mathbb{E} \left[\sum_{i=1}^n (X_i - Y)^2 \right] = n\sigma^2 + n\mu^2 - \sigma^2 - n\mu^2 = (n-1)\sigma^2$$

(Proof Complete)

Problem 2 [3 Points]

Statement: Let X and Y be independent random variables having geometric densities with parameters p_1 and p_2 respectively. Find $P(X > Y)$ and $P(X = Y)$. Let $q_1 = 1 - p_1$ and $q_2 = 1 - p_2$.

Part A: Find $P(X = Y)$ [1.5 Points]

Solution: Because X and Y are independent, $P(X = k, Y = k) = P(X = k)P(Y = k)$.

$$P(X = Y) = \sum_{k=1}^{\infty} P(X = k)P(Y = k) = \sum_{k=1}^{\infty} (q_1^{k-1}p_1) (q_2^{k-1}p_2)$$

$$P(X = Y) = p_1p_2 \sum_{k=1}^{\infty} (q_1q_2)^{k-1}$$

Let $i = k - 1$. As k goes from 1 to ∞ , i goes from 0 to ∞ :

$$P(X = Y) = p_1p_2 \sum_{i=0}^{\infty} (q_1q_2)^i$$

Using the sum of an infinite geometric series $\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$:

$$P(X = Y) = \frac{p_1p_2}{1 - q_1q_2} = \frac{p_1p_2}{1 - (1 - p_1)(1 - p_2)}$$

Part B: Find $P(X \geq Y)$ [1.5 Points]

Solution: Set up the joint probability sum over all cases where $X \geq Y$:

$$P(X \geq Y) = \sum_{j=1}^{\infty} P(Y = j)P(X \geq j)$$

The marginal probability $P(X \geq j)$ is the probability that the first $j - 1$ trials are failures:

$$P(X \geq j) = q_1^{j-1}.$$

Therefore,

$$P(X \geq Y) = \sum_{j=1}^{\infty} (q_2^{j-1}p_2) (q_1^{j-1})$$

Group the terms with the same exponent:

$$P(X \geq Y) = p_2 \sum_{j=1}^{\infty} (q_1q_2)^{j-1}$$

Let $i = j - 1$:

$$P(X \geq Y) = p_2 \sum_{i=0}^{\infty} (q_1q_2)^i$$

Using the sum of an infinite geometric series $\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$:

$$P(X \geq Y) = \frac{p_2}{1 - q_1q_2} = \frac{p_2}{1 - (1 - p_1)(1 - p_2)}$$